POLITECNICO DI MILANO

High Performance Computing Engineering

Quantum Physics – January 14, 2025

Please justify each answer and use clear handwriting. Define each symbol you employ. For each exercise, give the symbolic solution and substitute numerical values (if required) only at the end. Write your first name, family name, and personal code on each sheet in a readable way.

1. Consider a particle described by a two-level quantum system, with energy pure states $|0\rangle$ and $|1\rangle$ and energy eigenvalues $E_0 \neq 0$, and $E_1 = 0$, respectively. In such a system the pure states for the particle position along the x axis are $|x_0\rangle = c_0(i|0\rangle + |1\rangle)$, with eigenvalue $x_0 \neq 0$, and $|x_1\rangle = c_1(|0\rangle + \alpha|1\rangle)$, with eigenvalue $x_1 = 0$.

a) Determine the value that the α parameter in the expression of $|x_1\rangle$ must assume. Determine possible values of the constants c_0 and c_1 for the normalization of $|x_0\rangle$ and $|x_1\rangle$.

b) In energy representation, find the matrices corresponding to the Hamiltonian operator $\hat{H}$ and to the particle position operator $\hat{X}$.

c) Determine the time dependence of the position expectation value for a state which coincides with $|x_0\rangle$ at time $t = 0$.

d) For the same state as in point (c), evaluate the standard deviation of the particle position along x as a function of time.

2. In the ground state of a Helium (He) atom, the two electrons both occupy the $|1s\rangle$ single electron energy eigenstate. In the first excited state, one of the two electrons is promoted to the $|2s\rangle$ single electron energy eigenstate. The interaction between the two electrons can be neglected.

a) In energy representation and explicitly considering the spin of each electron, write all the kets corresponding to the ground state of the He atom. What is the ground state degeneracy (number of orthogonal states with the same energy)?

b) What is the maximum value of the *total* spin projection of the two electrons in the ground state of the He atom along a quantization axis of choice?

c) In energy representation and explicitly considering the spin of each electron, write all the kets corresponding to the first excited state of the He atom. What is the first excited state degeneracy (number of orthogonal states with the same energy)?

d) What is the maximum value of the *total* spin projection of the two electrons in the first excited state of the He atom along a quantization axis of choice?

Solution to the 1st exercise

a) Pure states corresponding to different values of an observable quantity must be orthogonal to each other:

$$\langle x_0|x_1\rangle = c_0^* c_1[(-i\langle 0| + \langle 1|)(|0\rangle + \alpha|1\rangle)] = c_0^* c_1[-i\langle 0|0\rangle - \alpha i\langle 0|1\rangle + \langle 1|0\rangle + \alpha\langle 1|1\rangle]$$
$$= c_0^* c_1(-i + \alpha) = 0.$$

The orthogonality condition is therefore satisfied for $\alpha = i$.
The normalization conditions for $|x_0\rangle$ and $|x_1\rangle$ are instead satisfied for:

$$\langle x_0|x_0\rangle = |c_0|^2[(-i\langle 0| + \langle 1|)(i|0\rangle + |1\rangle)] = |c_0|^2[\langle 0|0\rangle - i\langle 0|1\rangle + i\langle 1|0\rangle + \langle 1|1\rangle] = 2|c_0|^2 = 1$$
$$\langle x_1|x_1\rangle = |c_1|^2[(\langle 0| - i\langle 1|)(|0\rangle + i|1\rangle)] = |c_1|^2[\langle 0|0\rangle + i\langle 0|1\rangle - i\langle 1|0\rangle + \langle 1|1\rangle] = 2|c_1|^2 = 1$$

An arbitrary choice satisfying the normalization condition is $c_0 = c_1 = \frac{1}{\sqrt{2}}$.

b) By (arbitrarily) assigning $|0\rangle$ to $\begin{bmatrix}1\\0\end{bmatrix}$ and $|1\rangle$ to $\begin{bmatrix}0\\1\end{bmatrix}$, the matrix expression of the Hamiltonian operator is

$$\hat{H} = E_0|0\rangle\langle 0| + E_1|1\rangle\langle 1| = E_0|0\rangle\langle 0| = E_0\begin{bmatrix}1\\0\end{bmatrix}\cdot\begin{bmatrix}1 & 0\end{bmatrix} = \begin{bmatrix}E_0 & 0\\0 & 0\end{bmatrix}.$$

The matrix expression of the position operator in energy representation is instead:

$$\hat{X} = x_0|x_0\rangle\langle x_0| + x_1|x_1\rangle\langle x_1| = x_0|x_0\rangle\langle x_0| = \frac{x_0}{2}\begin{bmatrix}i\\1\end{bmatrix}\cdot\begin{bmatrix}-i & 1\end{bmatrix} = \frac{x_0}{2}\begin{bmatrix}1 & i\\-i & 1\end{bmatrix}.$$

c) The solution $|u(t)\rangle$ of the Schrödinger equation having $|x_0\rangle$ as the initial state at $t = 0$ is:

$$|u(t)\rangle = \frac{1}{\sqrt{2}}\left(ie^{-\frac{E_0}{\hbar}t}|0\rangle + |1\rangle\right) = \frac{1}{\sqrt{2}}\begin{bmatrix}i\,e^{-\frac{E_0}{\hbar}t}\\1\end{bmatrix}.$$

The position expectation value $\langle x\rangle$ as a function of time is then given by:

$$\langle x\rangle = \langle u(t)|\hat{X}|u(t)\rangle = \frac{x_0}{4}\begin{bmatrix}-i\,e^{\frac{E_0}{\hbar}t} & 1\end{bmatrix}\cdot\begin{bmatrix}1 & i\\-i & 1\end{bmatrix}\cdot\begin{bmatrix}i\,e^{-\frac{E_0}{\hbar}t}\\1\end{bmatrix} = \frac{x_0}{2}\left[1 + \cos\left(\frac{E_0}{\hbar}t\right)\right].$$

d) The standard deviation of the particle position is:

$$\sigma_x = \sqrt{\langle x^2\rangle - \langle x\rangle^2}.$$

Let us evaluate $\langle x^2\rangle$:

$$\langle x^2\rangle = \langle u(t)|\hat{X}^2|u(t)\rangle = \frac{x_0^2}{8}\begin{bmatrix}-i\,e^{\frac{E_0}{\hbar}t} & 1\end{bmatrix}\cdot\begin{bmatrix}1 & i\\-i & 1\end{bmatrix}\cdot\begin{bmatrix}1 & i\\-i & 1\end{bmatrix}\cdot\begin{bmatrix}i\,e^{-\frac{E_0}{\hbar}t}\\1\end{bmatrix} = \frac{x_0^2}{2}\left[1 + \cos\left(\frac{E_0}{\hbar}t\right)\right].$$

Therefor the standard deviation is:

$$\sigma_x = \sqrt{\langle x^2\rangle - \langle x\rangle^2} = \frac{x_0}{2}\left|\sin\left(\frac{E_0}{\hbar}t\right)\right|.$$

Solution to the 2nd exercise

a) There is only one ket for the He ground state that is compatible with the Pauli exclusion principle and the spin statistics theorem:

$$|\psi_0\rangle = \frac{1}{\sqrt{2}}|1s\rangle^{(1)}|1s\rangle^{(2)}\left(|\uparrow\rangle^{(1)}|\downarrow\rangle^{(2)} - |\downarrow\rangle^{(1)}|\uparrow\rangle^{(2)}\right),$$

with $\hat{s}_z^{(j)}|\uparrow\rangle^{(j)} = +\frac{\hbar}{2}|\uparrow\rangle^{(j)}$ and $\hat{s}_z^{(j)}|\downarrow\rangle^{(j)} = -\frac{\hbar}{2}|\downarrow\rangle^{(j)}$, where $\hat{s}_z^{(j)}$ acts on the jth electron and indicates the operator associated to the spin projection along the arbitrary z quantization axis. The $\frac{1}{\sqrt{2}}$ factor is for normalization purposes.

b) The maximum value of the total electron spin along the arbitrary z quantization axis for the He ground state is:

$$\left\langle\psi_0\right| \hat{s}_z^{(1)} + \hat{s}_z^{(2)}\left|\psi_0\right\rangle = 0.$$

c) According to the spin statistics theorem there are four energy degenerate and orthogonal states that could be associated to the first excited state of the He atom. A possible choice is the following:

$$|\psi_1\rangle = \frac{1}{\sqrt{2}}\left(|1s\rangle^{(1)}|2s\rangle^{(2)} - |2s\rangle^{(1)}|1s\rangle^{(2)}\right)|\uparrow\rangle^{(1)}|\uparrow\rangle^{(2)},$$
$$|\psi_2\rangle = \frac{1}{\sqrt{2}}\left(|1s\rangle^{(1)}|2s\rangle^{(2)} - |2s\rangle^{(1)}|1s\rangle^{(2)}\right)|\downarrow\rangle^{(1)}|\downarrow\rangle^{(2)},$$
$$|\psi_3\rangle = \frac{1}{\sqrt{2}}\left(|1s\rangle^{(1)}|\uparrow\rangle^{(1)}|2s\rangle^{(2)}|\downarrow\rangle^{(2)} - |2s\rangle^{(1)}|\downarrow\rangle^{(1)}|1s\rangle^{(2)}|\uparrow\rangle^{(2)}\right),$$
$$|\psi_4\rangle = \frac{1}{\sqrt{2}}\left(|1s\rangle^{(1)}|\downarrow\rangle^{(1)}|2s\rangle^{(2)}|\uparrow\rangle^{(2)} - |2s\rangle^{(1)}|\uparrow\rangle^{(1)}|1s\rangle^{(2)}|\downarrow\rangle^{(2)}\right).$$

The $\frac{1}{\sqrt{2}}$ factor is for normalization purposes. Please note that $|\psi_1\rangle$ and $|\psi_2\rangle$ are eigenstates for *both* $\hat{s}_z^{\text{tot}} = \left(\hat{s}_z^{(1)} + \hat{s}_z^{(2)}\right)$ and $(\hat{\mathbf{s}}^{\text{tot}})^2 = \left(\hat{\mathbf{s}}^{(1)} + \hat{\mathbf{s}}^{(2)}\right)^2$, with eigenstates equal to $\pm\hbar$ and $2\hbar^2$, respectively. Conversely, $|\psi_3\rangle$ and $|\psi_4\rangle$ are eigenstates of $\hat{s}_z^{\text{tot}}$, with eigenvalue 0, but *not* of $(\hat{\mathbf{s}}^{\text{tot}})^2$. Eigenvectors for *both* $\hat{s}_z^{\text{tot}}$ *and* $(\hat{\mathbf{s}}^{\text{tot}})^2$ would be $\frac{1}{\sqrt{2}}(|\psi_3\rangle + |\psi_3\rangle)$ and $\frac{1}{\sqrt{2}}(|\psi_3\rangle - |\psi_3\rangle)$, with $(\hat{\mathbf{s}}^{\text{tot}})^2$ eigenvalues equal to $2\hbar^2$ and 0, respectively.

d) The maximum value of the total electron spin along the arbitrary z quantization axis for the He ground state is:

$$\left\langle\psi_1\right| \hat{s}_z^{(1)} + \hat{s}_z^{(2)}\left|\psi_1\right\rangle = \hbar.$$